

Signaloid C0-microSD Starter UxHw® Module

Description

The Signaloid C0-microSD Starter UxHw® Module is a low-power hot-swappable system-on-module (SoM) that brings Signaloid’s distribution-extended compute hardware (UxHw®) technology to any system with a microSD slot. Host systems into which it is plugged treat it as a standard SD-compatible block storage device over the 4-wire SD protocol. As a result, adding it to an existing system does not require circuit board redesign or host firmware changes.

Features

- Programmable in C with the Signaloid UxHw Toolchain, based on clang 17.0. Flash binaries to the built-in Signaloid C0 RISC-V-based SoC over the SD interface.
- Run sensor fusion algorithms which can exploit UxHw technology, to achieve more than 5× speedup over comparable solutions.
- Appears as a regular block device on any operating system that supports the standard SD 1.x protocol, with no additional drivers required.
- Supports SD-over-SPI protocol for easy integration with low-cost microcontrollers.
- Fits standard microSD slots at 0.8 mm thickness.
- Power dissipation of less than 30 mW allows it to be used for compute offload without additional cooling.

Modes of Operation

- The C0-microSD has three distinct modes of operation:
- **Bootloader** mode: Flash new custom bitstreams, update the firmware of the C0-microSD, and flash new user data to the non-volatile memory.
 - **Signaloid SoC** mode: Load and run C/C++ applications on the Signaloid C0 core, optionally taking advantage of UxHw technology in your algorithms.
 - **Custom User Bitstream** mode: Load custom user bitstreams onto the device.

Compatibility

- Programmable logic controllers (PLCs) that have unused microSD slots.
- Android mobile devices that have a microSD slot¹.
- Microcontroller-based embedded systems that support full SD 1.x or SD over SPI.
- Single-board computers with unused microSD slots.



Target Applications

- State estimation algorithms (e.g., Kalman filters and particle filters) for robotics applications such as autonomous mobile robots (AMRs).
- Bayesian decision making over collected positive and negative event samples.
- Battery state of charge estimation in battery energy storage systems (BESSs), electric vehicles, and portable electronic devices.

Table of Contents

Description.....	1
Features.....	1
Modes of Operation.....	1
Compatibility.....	1
Target Applications.....	1
Table of Contents.....	1
Technical Specifications.....	2
Typical Application Circuits.....	2
Functional Block Diagram.....	2
Package Diagram.....	3
Pinout Diagram.....	3
Pinout Description.....	3
Benchmarks.....	4
Public Code Examples.....	4
C0-microSD LiteX integration.....	4
Further Information.....	4
Revision History.....	4

¹ Requires root privileges.

Technical Specifications

Signaloid C0-microSD	Technical data
Signaloid C0 RISC-V-based SoC	- RV32I instruction set 12 MHz clock frequency 128 KiB on-chip memory
Lattice Semiconductor iCE40UP5K FPGA	5280 logic cells 120 Kb (15 KiB) dual-port block RAM 1 Mb (128 KiB) single-port RAM
Non-volatile memory	Renesas AT25QL128A 128 Mb (16 MiB) SPI Flash
GPIO pin count	Total of 13 GPIOs: 2 on-board LEDs 6 GPIOs routed to pads on the microSD interface 5 GPIOs on test pads
UxHw Capabilities	
Maximum representation size	TTR16
Number of in-flight distributions	600 (TTR8) 288 (TTR16)
I/O minimum latency (read/write) ²	0.8 ms / 6.0 ms
I/O maximum throughput	100 Mbit/s
Electrical Specifications	
Supply voltage (V_{DD})	3.14 V–3.5 V
I/O Supply voltage (V_{DDIO})	LVC MOS 3.3: 3.13 V–3.46 V LVC MOS 1.8: 1.71 V–1.89 V
I/O max rating	LVC MOS 3.3: 8 mA LVC MOS 1.8: 4 mA
Current consumption ³	Minimum: 4.96 mA Average: 5.18 mA Maximum: 7.43 mA
Miscellaneous	
Operating temperature	-40 °C–85 °C
Dimensions	11 mm × 15 mm × 0.8 mm
Typical operating temperature ⁴	< 35 °C

Table 1: C0-microSD technical specifications

² Measured on transactions of 512-byte blocks using the **dd** command.

³ Using 3.3V supply and adding two uniformly-distributed double-precision floating-point values (3.0 ± 0.6 and 5.0 ± 0.6) using UxHw technology.

⁴ Measured after 30 minutes of continuous operation running a typical workload in an open-lid microSD slot and 24 °C ambient temperature.

Typical Application Circuits

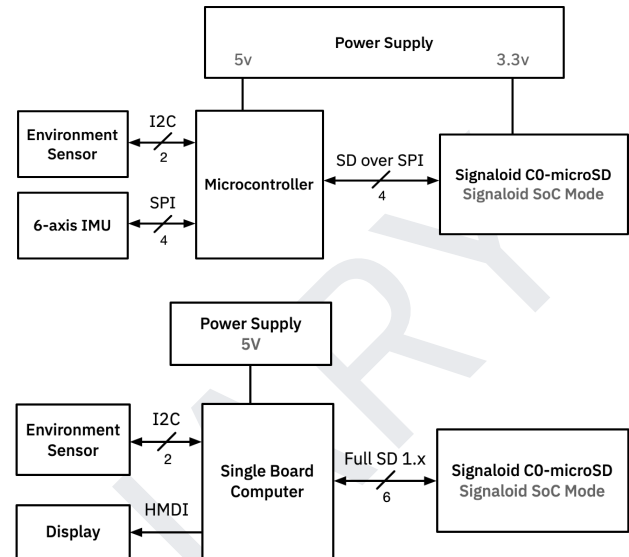


Figure 1: Example circuit of the Signaloid C0-microSD acting as a co-processor on a microcontroller-based embedded system (top), or on a single-board computer (bottom).

Functional Block Diagram

The Signaloid C0-microSD is based on the iCE40 FPGA from Lattice Semiconductor. In addition to the iCE40 FPGA, the Signaloid C0-microSD SOM contains 128Mbit of serial NOR flash for storing bitstreams, firmware, and user data. The device also features two on-board LEDs, one red and one green, connected to the FPGA.

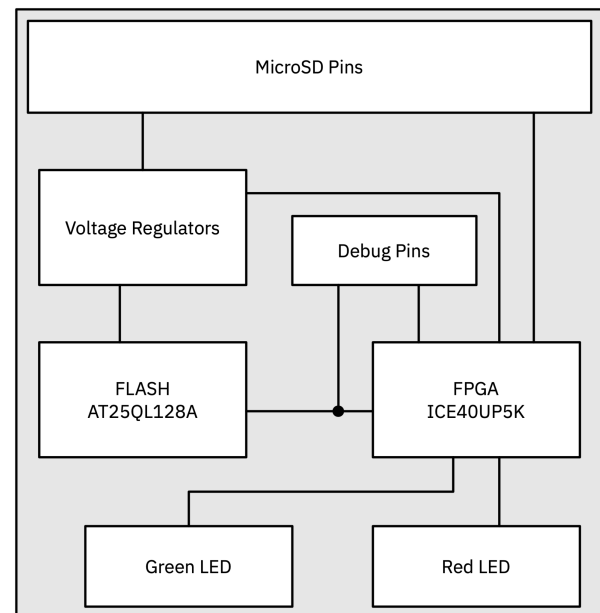


Figure 2: C0-microSD functional diagram.

Package Diagram

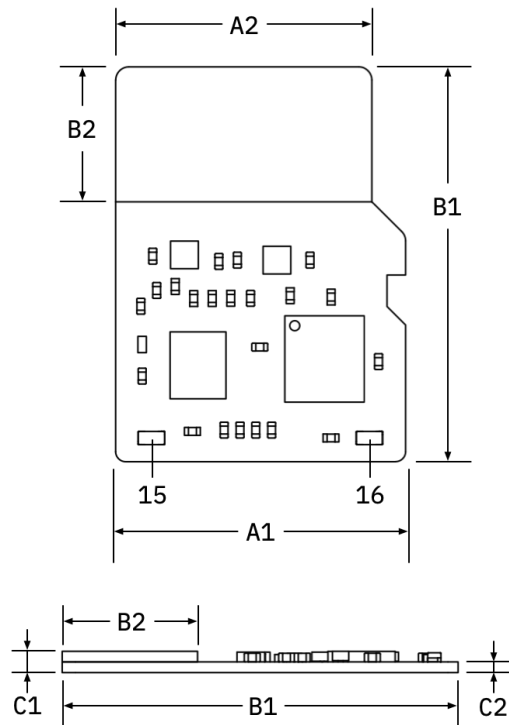


Figure 3: C0-microSD package diagram

Dimension	Typical	Tolerance	Unit
A1	11	± 0.2	mm
A2	9.7	± 0.2	mm
B1	15	± 0.2	mm
B2	9.9	± 0.2	mm
C1	0.8	± 0.2	mm
C2	0.4	± 0.2	mm

Table 2 : C0-microSD dimensions

Pinout Diagram

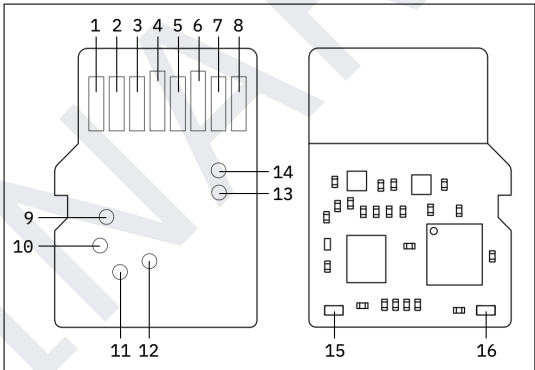


Figure 4 : C0-microSD pinout diagram

Pinout Description

Number	Pin Name	FPGA Pin	Standard	Description
1	SD_DAT2	E5	LVC MOS 3.3	GPIO pin connected to the SD bus DAT2
2	SD_DAT3	F5	LVC MOS 3.3	GPIO pin connected to the SD bus DAT3
3	SD_CMD	A4	LVC MOS 3.3	GPIO pin connected to the SD bus CMD
4	VDD	-	-	Supply voltage
5	SD_CLK	B3	LVC MOS 3.3	GPIO pin connected to the SD bus CLK
6	VSS	-	-	Supply ground
7	SD_DAT0	A1	LVC MOS 3.3	GPIO pin connected to the SD bus DAT0
8	SD_DAT1	A2	LVC MOS 3.3	GPIO pin connected to the SD bus DAT1
9	CONFIG_SCLK	D1	LVC MOS 1.8	Clock pin of SPI configuration port
10	CONFIG_MISO	E1	LVC MOS 1.8	MISO pin of SPI configuration port
11	CONFIG_MOSI	F1	LVC MOS 1.8	MOSI pin of SPI configuration port
12	CONFIG_CSn	C1	LVC MOS 1.8	Chip select of SPI configuration port (active low)
13	COFIG_CRESETn	F3	LVC MOS 1.8	FPGA reset pin (active low)
14	CONFIG_DONE	D3	LVC MOS 1.8	Configuration Done pin
15	GREEN_LED	A5	LVC MOS 3.3 Open Drain	Green status LED
16	RED_LED	B5	LVC MOS 3.3 Open Drain	Red status LED

Table 3 : C0-microSD pinout

Benchmarks

The benchmarks below evaluate the speedup achieved by replacing Monte-Carlo-based uncertainty quantification of sensor calibration and conversion routines, with UxHw-based computation running on the C0-microSD. The evaluations compare Monte Carlo executions on a RP2350 microcontroller (ARM Cortex-M33), running at 150 MHz and dissipating about 39mW, to computation implemented using UxHw configured with the TTR16 representation running on the C0-microSD. The UxHw representation is chosen such that the output probability distribution has the same degree of convergence as the Monte Carlo result. The application kernels in the table correspond to the manufacturer-specified calibration algorithms that convert each measured ADC value into a calibrated and usable sensor output.

Application	Monte Carlo Iterations	Monte Carlo Latency	UxHw Latency ⁵	UxHw Speedup
Flusso FLS110	291k	3000 ms	1630 ms	1.8×
NXP MPX4100A	185k	700 ms	420 ms	1.6×
NXP MPX6250A	215k	800 ms	420 ms	1.9×
Sensirion SDP3x	193k	2300 ms	420 ms	5.5×
Sensirion SFM3100	2.7M	8200 ms	23.56 ms	350×
Sensirion SHT3xARP	193k	1600 ms	1260 ms	1.3×
Sensirion SHT4xI	204k	1700 ms	1260 ms	1.4×

Table 4 : C0-microSD UxHw speedup of sensor calibration applications compared to Monte-Carlo-based implementations using the RP2350 microcontroller.

Public Code Examples

C0-microSD LiteX integration

Use the Signaloid C0-microSD as an FPGA development board and configure it with your custom SoC implementation using LiteX to access sensors and displays using I2C.



<http://github.com/signaloid/Signaloid-C0-microSD-litex-I2C-demo>

For more information, contact sales@signaloid.com.

Revision History

Revision 1.1, July 2026

Section	Change Summary
All	Updated formatting
Features	Updated wording
Compatibility	Introduced compatibility section
Target applications	Introduced target applications section
Benchmarking	Introduced benchmarking data of sensor example applications.

Revision 1.0, February 2025

Section	Change Summary
All	Initial Release

⁵ Latency for achieving the same degree of convergence of the distribution result as the multi-iteration Monte Carlo.